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Level 200 • Second Semester • Handout

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES

EFFURUN

SEMINAR

ON

ORDINARY DIFFERENTIAL EQUATION AND PARTIAL
DIFFERENTIAL EQUATION.

FROM THE DEPARTMENT OF MATHEMATICS.

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ABSTRACT

Ordinary Differential Equations (ODEs) and Partial Differential Equations (PDEs) represent two foundational pillars of mathematical modeling, each with its distinct characteristics and applications.

This abstract delves into the fundamental aspects and practical applications of ODEs and PDEs in various

fields, shedding light on their crucial roles in understanding dynamic systems.

ODEs, which involve single independent variables, are pivotal in modeling dynamic processes that evolve

over time. From the motion of celestial bodies to the growth of populations, ODEs provide essential tools for predicting future behavior based on initial conditions and governing equations. Their simplicity

and versatility make ODEs indispensable in fields like physics, engineering, and biology.

On the other hand, PDEs, which incorporate multiple independent variables, are adept at capturing complex spatial and temporal variations. They find extensive use in modeling phenomena such as heat

diffusion, fluid dynamics, and electromagnetic propagation. PDEs enable us to analyze how systems change in response to various factors, making them indispensable in the study of mathematical physics,

engineering, and environmental science.

This abstract also highlights the interdisciplinary nature of ODEs and PDEs, emphasizing their applications in diverse domains, from predicting stock market trends to simulating climate patterns.

Furthermore, it underscores their role in the advancement of mathematical theory, contributing to a

deeper understanding of the mathematical underpinnings of the natural world.

In conclusion, ODEs and PDEs serve as powerful tools in mathematical modeling, enabling us to comprehend and predict the behavior of dynamic systems across a wide spectrum of disciplines. This

abstract provides a glimpse into the significance and versatility of these equations, showcasing their

profound impact on both theoretical mathematics and real-world problem-solving.

CHAPTER ONE

1.0 INTRODUCTION

In Mathematics, a differential equation is an equation that contains one or more functions with its derivatives. The derivatives of the function define the rate of change of a function at a point. It is mainly

used in fields such as physics, engineering, biology and so on.

The primary purpose of the differential equation is the study of solutions that satisfy the equations and the

properties of the solutions.

1.1 DIFFERENTIAL EQUATIONS.

A differential equation is an equation which contains one or more terms and the derivatives of one variable (i.e. dependent variable) with respect to the other variable (i.e. independent variable)

$$dy/dx = f(x)$$

Here “x” is an independent variable and “y” is a dependent variable

For example, $dy/dx = 5x$

A differential equation contains derivatives which are either partial derivatives or ordinary derivatives. The derivative represents a rate of change, and the differential equation describes a relationship between the quantity that is continuously varying with respect to the change in another quantity.

Order of Differential Equation.

The order of the differential equation is the order of the highest order derivative present in the equation.

Here some examples for different orders of the differential equation are given.

$Dy/dx = 3x + 2$, The order of the equation is 1

$(d^2y/dx^2) + 2(dy/dx) + y = 0$. The order is 2

$(dy/dt) + y = kt$. The order is 1

First Order Differential Equation.

You can see in the first example, it is a first-order differential equation which has degree equal to 1. All

the linear equations in the form of derivatives are in the first order. It has only the first derivative such as

dy/dx , where x and y are the two variables and is represented as:

$$dy/dx = f(x, y) = y'$$

Second-Order Differential Equation.

The equation which includes the second-order derivative is the second-order differential equation. It is

represented as;

$$d/dx(dy/dx) = d^2y/dx^2 = f''(x) = y''$$

Degree of Differential Equation

The degree of the differential equation is the power of the highest order derivative, where the original

equation is represented in the form of a polynomial equation in derivatives such as y' , y'' , y''' and so on.

Suppose $(d^2y/dx^2) + 2(dy/dx) + y = 0$ is a differential equation, so the degree of this equation here is 1. See

some more examples here:

$$Dy/dx + 1 = 0, \text{ degree is } 1$$

$$(y''')^3 + 3y'' + 6y' - 12 = 0, \text{ degree is } 3$$

$(dy/dx) + \cos(dy/dx) = 0$; it is not a polynomial equation in y' and the degree of such a differential equation can not be defined.

Note:

Order and degree (if defined) of a differential equation are always positive integers.

Types of Differential Equations

Differential equations can be divided into several types namely

1. Ordinary Differential Equations
2. Partial Differential Equations
3. Linear Differential Equations
4. Nonlinear differential equations
5. Homogeneous Differential Equations
6. Nonhomogeneous Differential Equations.

In this chapter, we shall be considering 1 and 2 only.

Ordinary Differential Equations (ODE)

An ordinary differential equation involves functions and its derivatives. It contains only one independent variable and one or more of its derivatives with respect to the variable. Often, our goal is to

solve an ODE, i.e., determine what function or functions satisfy the equation. Often, our goal is to solve

an ODE, i.e., determine what function or functions satisfy the equation.

If you know what the derivative of a function is, how can you find the function itself? You need to find

the antiderivative, i.e., you need to integrate. For example, if you are given

$$Dxdt(t) = \cos(t)$$

Then what is the function $x(t)$?

Since the antiderivative of $\cos(t)$ is $\sin(t)$, then $x(t)$ must be $\sin(t)$

Except we forgot one important point: there is always an arbitrary constant that we cannot determine if

we only know the derivative. Therefore, all we can determine from the above equation is that

$X(t) = \sin(t) + C$ for some arbitrary constant C . You can verify that indeed $x(t)$ satisfies the equation

$$dx/dt = \cos(t).$$

In general, solving an ODE is more complicated than simple integration. Even so, the basic principle is

always integration, as we need to go from derivative to function. Usually, the difficult part is determining

what integration we need to do.

Types of Ordinary Differential Equations

The Ordinary Differential Equations can be of various types, and their application is in different mathematics areas. We can primarily classify ODE or Ordinary Differential Equations into three types.

They are stated below:

1. Autonomous Ordinary Differential Equation
2. Linear Ordinary Differential Equations
3. Non-linear Ordinary Differential Equations
- 4.

- What are Autonomous (ODE) Ordinary Differential Equations?

A differential equation that is not dependent on the variable, such as x , is known as an autonomous differential equation.

Example: $dy/dv = by + 9$

- What are Linear (ODE) Ordinary Differential Equations?

Those differential equations that can be written as the linear combinations of the derivatives of, say y ,

then they are known as linear ordinary differential equations. An example of Linear Ordinary Differential

Equation is $dy/dx + 2y = \sin(x)$.

- What are Non-linear (ODE) Ordinary Differential Equations?

Those differential equations that cannot be written in the style of linear combinations of the derivatives

of, say y , then they are known as a non-linear ordinary differential equation. An example of Linear Ordinary Differential Equation is $y'' = f(x, y)$.

Partial Differential Equations. (PDE)

A Partial Differential Equation commonly denoted as PDE is a differential equation containing partial derivatives of the dependent variable (one or more) with more than one independent variable.

The PDE is said to be linear if f is a linear function of u and its derivatives. The simple PDE is given by;

$$\partial u / \partial x (x, y) = 0$$

The above relation implies that the function $u(x, y)$ is independent of x which is the reduced form of partial differential equation formula stated above. The order of PDE is the order of the highest derivative

term of the equation.

Partial Differential Equation Types

The different types of partial differential equations are:

1. First-order Partial Differential Equation
2. Linear Partial Differential Equation
3. Quasi-Linear Partial Differential Equation
4. Homogeneous Partial Differential Equation

Let us discuss these types of PDEs here.

- First-Order Partial Differential Equation

In Maths, when we speak about the first-order partial differential equation, then the equation has only the

first derivative of the unknown function having 'm' variables. It is expressed in the form of;

$$F(x_1, \dots, x_m, u, u_{x_1}, \dots, u_{x_m}) = 0$$

- Linear Partial Differential Equation

If the dependent variable and all its partial derivatives occur linearly in any PDE then such an equation is

called linear PDE otherwise a nonlinear PDE.

- Quasi-Linear Partial Differential Equation

A PDE is said to be quasi-linear if all the terms with the highest order derivatives of dependent variables

occur linearly, that is the coefficient of those terms are functions of only lower-order derivatives of the

dependent variables. However, terms with lower-order derivatives can occur in any manner.

- Homogeneous Partial Differential Equation

If all the terms of a PDE contain the dependent variable or its partial derivatives then such a PDE is called

non-homogeneous partial differential equation or homogeneous otherwise.

SOCIAL LIFE AND PHYSICAL LIFE.

SOCIAL LIFE.

Social life refers to the interactions, relationships, and activities that individuals engage in within a society or community. It encompasses various aspects such as friendships, family dynamics, social events, and cultural practices. Social life plays a crucial role in shaping our identity, providing support,

and fostering a sense of belonging.

Now, let's define some terms:

- Social interactions: These are the exchanges and communications between individuals, which can occur in various settings such as work, school, or social gatherings. They play a vital role in building relationships and fostering social connections.
- Relationships: These refer to the connections and bonds between individuals, whether they are romantic, familial, or friendships. Relationships are built on trust, mutual understanding, and shared experiences.
- Activities: These are the actions and engagements individuals participate in, which can include hobbies, sports, volunteering, or attending events. Activities contribute to personal growth,

enjoyment, and socialization.

PHYSICAL LIFE.

Physical life refers to the physical well-being and the activities of an individual. It includes aspect such as physical health, exercise, nutrition and overall bodily functions. Physical life is essential for maintaining a healthy body, preventing illness and promoting overall well-being.

Now, let's define some terms:

- **Physical health:** This refers to the overall well-being of an individual's body, including factors such as fitness, nutrition, and the absence of illness or disease. Maintaining good physical health is crucial for optimal functioning and longevity.
- **Exercise:** This involves physical activity that is structured and planned, aimed at improving fitness, strength, and endurance. Regular exercise has numerous health benefits, including reducing the risk of chronic diseases and improving mental well-being.
- **Nutrition:** This refers to the intake of food and nutrients necessary for the body's growth, development, and maintenance. A balanced and nutritious diet is essential for overall health and vitality.

In summary, social life encompasses interactions, relationships, and activities within a society, while

physical life focuses on the well-being and activities related to the body. Understanding and nurturing

both aspects are important for a fulfilling and balanced life.

CHAPTER 2

APPLICATION OF ODE AND PDE ON SOCIAL LIFE.

Ordinary differential equation and partial differential equation have various application in understanding

physical phenomena in our daily lives. Below, we shall look at some specific examples of the application

of ODE and PDE to human life and life in general.

1. Accumulation of Wealth.

Suppose that you now have \$10,000 and you expect to save an additional \$4,000 during each year, and all

of this is deposited in a bank paying 5% interest compounded continuously. Let $y(t)$ be your bank balance

in thousands of dollars after t years.

- Write a differential equation and initial condition to model your bank balance
- Solve your differential equation and initial condition.
- Use your solution to find your bank balance after 10 years.

Solution.

- Required differential equation is:

$$y' = 0.05y + 4, y(0) = 10$$

- To solve with this first order differential equation, we can use variable separation method.

$$y' = 0.05y + 4$$

11

11

$$= (0.05\pi + 4)$$

$$x_1 = (0.05x_1 + 4)x_1$$

□□□□□ □□□ h □□□□□ □□ (0.05□ + 4)

1

$$\square\square = \square\square$$

$0.05x + 4$

□□□□□□□□□□ □□ h □□□□□

1

$$\int_{0.05+4}^{\infty} f(x) dx = \int_{\infty}^{\infty} f(x) dx$$

1 0.05

0.05

$$\int_{0.05}^{+4} f(x) dx = \int_{0.05}^{+4} f(x) dx$$

1

$$\ln(0.05x + 4) = \square + \square 1$$

0.05

$$\ln(0.05x + 4) = 0.05x + 1$$

Taking the exponent of both sides

$$\ln(0.05x+4) = 0.05x + 1$$

$$0.05 \square + 4 = \square 0.05 \square \square \square 1$$

$$0.05x + 4 = 0.05x + 2$$

$$0.05\% = 2\% - 0.05\% - 4\%$$

□□□□□ □ h □□□□□□ □□ □□□□□□, □□ h □□□ □ h □□

1

$$\sigma = [\sigma \quad 0.05\sigma - 4]$$

0.05 2

$$\sigma = 20(\sigma^2 - 0.05\sigma - 4)$$

$$\therefore \pi = 20 - 2(0.05) - 80$$

□□□□, □ h □□□□□□ □□□□□□ □□ □ h □□□□□□□□□□ □□□□□□ □□:

$$\pi = 20\pi^2 - 0.05\pi - 80$$

□□□□ □ h □□□□□□ □□□□□□□ □(□) = □(0) = 10, □ h □□□ □ = 0, □□ □□□ □2

$$\pi = 20\pi^2 - 0.05\pi - 80$$

$$10 = 20 - 2 - 0.05(0) - 80$$

$$10 = 20 \square 2 \square 0 - 80$$

$$10 = 20 \cdot 2 - 80$$

$$10 + 80 = 20 \times 2$$

$$90 = 20 \sqcup 2$$

90

$$\sigma^2 = 4.5$$

20

2. A population of rabbits on an isolated island is experiencing continuous exponential growth. The initial population is 100 rabbits, and the population doubles every year. Write a first order differential equation to model this population growth, and find its solution. How many rabbits are expected to be on the island after 5 years?

Solution:

Step 1- Formulating the differential equation:

Let $P(t)$ be the population of rabbits at time t . Since the population doubles every year, the growth rate is proportional to the current population, and we can write the differential equation as:

$$\frac{dP}{dt} = kP$$

Where k is the growth constant.

Step 2- Solving the differential equation:

This is a separable differential equation. We can separate variables and integrate:

1

$$\frac{dP}{P} = k dt$$

2

Integrating:

1

$$\int \frac{dP}{P} = \int k dt$$

$$\ln P = kt + C$$

Taking the exponential of both sides

$$P = e^{kt+C}$$

$$P = e^C (e^{kt}) \dots \dots \dots \text{ii}$$

We can now find the value of C using the initial condition;

$$P(0) = 100$$

i.e $P = 100$ at $t = 0$; given

$$P = e^C (e^{kt})$$

$$100 = e^C [e^{k(0)}]$$

$$100 = e^C$$

Taking the natural logarithm of both sides;

$$\ln 100 = \ln e^C$$

$$\ln 100 = C \ln e$$

$$\ln 100 = C(1)$$

$$C = \ln 100$$

Substituting into equation(ii)

$$P = e^{kt+C} (100)$$

$$P = 100 e^{kt} \quad \text{..... iii}$$

$$P = 100 e^{kt}$$

Step 3- Finding the population after 5years:

We want to find $p(5)$ where $t = 5$:

Since we know that the population doubles every year, we can use this information to find k :

$$2 = e^{kt}$$

$$\ln 2 = \ln e^{kt}$$

$$\ln 2 = k \ln e$$

$$\ln 2 = k$$

$$K = \ln 2$$

Substituting k into (iii) at $t = 5$

$$P(5) = 100 e^{kt}$$

$$= 100 e^{(\ln 2)(5)}$$

$$= 100 e^{5 \ln 2}$$

$$5$$

$$= 100 e^{\ln 2^5}$$

$$= 100 (2^5)$$

$$P(5) = 3200$$

So they are expected to be 3200 rabbits in the island after 5years.

COURSE OF STUDY.

In my course of study, I have also encountered the application of ODEs and PDEs in various fields. For example, in physics, ODEs and PDEs are used to model the behavior of particles and waves. In engineering, they are applied to analyze and design systems such as control systems and signal processing. Overall, ODEs and PDEs are fundamental tools in understanding and solving problems in the physical sciences and engineering disciplines.

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